

## Chapter-4

# MOTION IN ONE DIMENSION

$$\frac{\text{Distance}}{\text{Displacement}} \geq 1; \frac{\text{Speed}}{\text{Velocity}} \geq 1$$

$$\text{Avg. speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$\text{Avg. velocity} = \frac{\text{Total displacement}}{\text{Total time}}$$

- If a body moves certain distance with speed  $V_1$  and returns to same point with speed  $V_2$  then,

$$\text{Average speed; } \langle V \rangle = \frac{2V_1V_2}{V_1 + V_2}$$

- If a person moves with equal distances with speed  $V_1, V_2, V_3$  and so on then average speed

$$\langle V \rangle = \frac{1}{V_1} + \frac{1}{V_2} + \frac{1}{V_3} + \dots + \frac{1}{V_n}$$

$$\text{Avg. speed } \langle V \rangle = \text{H.M. of } V_1, V_2, V_3, \dots, V_n$$

- If a body moves with different speed  $V_1, V_2, V_3, \dots, V_n$  in equal time interval then, average speed

$$\langle V \rangle = \frac{V_1 + V_2 + \dots + V_n}{n} = \text{A.M.}$$

- Equation of kinematics are applicable for constant acceleration. i.e. when acceleration is not varying with time.
- If the  $x - t$  graph is a straight line parallel to  $x$ -axis then the body is at rest.
- The straight line inclined to time axis in  $x - t$  graph represents constant velocity.
- In  $x - t$  graph the straightline inclined to time axis at an angle greater than  $90^\circ$ , show negative velocity.
- No line in  $x - t$  graph can be perpendicular to time axis because it will represent infinite velocity.
- If the  $x - t$  graph is a curve whose slope decreases continuously with time, then the velocity of the body goes on decreasing continuously and the motion of the body is retarded.
- If the  $x - t$  graph is a curve whose slope continuously increases, then the velocity of the body is continuously increasing and the body is accelerated.

- If the  $v - t$  graph is a straight line parallel to time axis, then the acceleration of the body is zero.
- If the  $v - t$  graph is a straight line inclined to time axis with positive slope, then that body is moving with constant acceleration.
- If  $v - t$  graph is a straight line inclined to time axis with negative slope, then the body is retarded.
- The velocity and acceleration of a body need not be in same direction.
- Velocity and acceleration of a body need not be zero simultaneously.
- A body in equilibrium has **zero acceleration** only. All other quantities need not be zero.
- If a body travels with a uniform acceleration  $a_1$  for a time interval  $t_1$  and with uniform acceleration  $a_2$  for a time interval  $t_2$ , then the average acceleration  $a = \frac{a_1t_1 + a_2t_2}{t_1 + t_2}$
- For a body moving with uniform acceleration, the average velocity =  $\frac{(u + v)}{2}$ , where  $u$  is the initial velocity and  $v$  is the final velocity.
- The distance travelled by the body in successive seconds is in the ratio  $1 : 3 : 5 : 7 \dots$  etc.
- When the body is starting from rest, the distances travelled by the body in the first second, first two seconds, first three seconds, ..... etc. are in the ratio of  $1 : 4 : 9 : 16 : 25 \dots$  etc.
- When a body is dropped freely from the top of the tower and another body is projected horizontally from the same point, both will reach the ground at the same time.
- If velocity-time graph is a curve whose slope decreases with time, then the acceleration goes on decreasing.
- If the  $v - t$  graph is a curve whose slope increases with time, the acceleration of the body goes on increasing.

# Motion In one Dimension

- The  $v - t$  graph normal to time axis is not a practical possibility because it means that the acceleration of the body is infinite.
- The area between  $\vec{a} - t$  graph and the time axis gives change in velocity.
- If the  $a - t$  graph is a straight line with positive slope, then the acceleration is continuously increasing if the slope is negative, the acceleration is continuously decreasing.
- If the particle starts from rest and the distance covered by it in time  $t$  be  $s$ , then
  - (i) if  $s \propto t$ , the acceleration is zero
  - (ii) if  $s \propto t^2$ , the acceleration is constant
  - (iii) if  $s \propto t^3$ , the acceleration varies as the time ( $a \propto t$ )
- If the distance covered ( $s$ ) by a particle is proportional to  $t^{3/2}$ , then the power dissipated by it is constant.
- If  $\vec{v}$  makes certain angle with  $\vec{a}$ , then the path of the particle is a parabola.
- Speed is always a positive quantity, however it may increase or decrease with time.
- For uniform motion: (a) Distance covered = magnitude of displacement; (b) The motion is along a straight line; (c) Direction of motion does not change.
- If the angle between  $\vec{a}$  and  $\vec{v}$  is between  $90^\circ$  and  $270^\circ$ , then the motion is retarded. While if the value of  $\theta$  lies between  $-90$  and  $90^\circ$ , then the motion is accelerated.
- If  $\vec{a}$  and  $\vec{v}$  are along same straight line, then the path of the particle is a straight line.
- If  $\vec{a}$  and  $\vec{v}$  are along different directions then the path of the particle is curved (e.g. projectile motion and circular motion)
- If air resistance is not neglected then  
time of descent  $>$  time of ascent
- At highest point,  
 $a_{\text{eff}} = g$
- If the body is thrown upward then it will rise until its vertical velocity becomes zero. Maximum height attained is  $h = \frac{u^2}{2g}$ .
- If air resistance is negligible then the time of rise is equal to time of fall, each is equal to  $t = \frac{u}{g}$ .

- The body returns to the straight point with the same velocity with which it was thrown.
- If a packet is dropped at a height 'h', from an aeroplane or balloon ascending with a velocity  $u$ , then, the time taken by the packet to reach ground is given by

$$t = \frac{u}{g} + \sqrt{\left(\frac{u}{g}\right)^2 + \frac{2h}{g}}$$

- If a person is throwing balls into air after another, he throws the second ball when first ball is at highest point and he throws 'n' ball every second, then maximum height attained by ball is  $h = \frac{g}{2n^2}$ .

## Relative velocity

- (i)  $\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$
  - (ii)  $\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$
- If two bodies are moving in same direction then  
 $|\vec{V}_{AB}| = (V_A - V_B)$
  - If two bodies are moving in opposite direction then  
 $|\vec{V}_{AB}| = (V_A + V_B)$
  - If two bodies are moving perpendicularly then,  
 $|\vec{V}_{AB}| = \sqrt{V_A^2 + V_B^2}$
  - If rain drops are falling vertically with a velocity  $v$  and a person is walking horizontally with a velocity  $u$ , then he should hold an umbrella at an angle  $\theta$  with vertical given by  $\tan\theta = \frac{u}{v}$ , to prevent himself from being wet.
  - A boat moving with a velocity  $v$  in still water crosses a river which is flowing with a velocity  $u$ , then:
    - (i) To reach the opposite bank in minimum time, the boat must move at right angles to the current and time taken to cross the river.  
 $t = \frac{D}{v}$ , where  $D$  is width of river.
    - (ii) To go straight across to the opposite bank, the boat must move at an angle  $\theta = \sin^{-1}\left(\frac{u}{v}\right)$  with the vertical or  $\left[90^\circ + \sin^{-1}\frac{u}{v}\right]$  with the direction of current and time.  
 $t = \frac{D}{\sqrt{v^2 - u^2}}$

## Multiple Choice Questions

1. A stationary man observes that the rain is falling vertically downward. When he starts running with a velocity 12km/h he observes that the rains is

falling at an angle  $60^\circ$  with the vertical. The actual velocity of rain is:

- a.  $12\sqrt{3}$  km/h      b. 12 km/h  
c.  $4\sqrt{3}$  km/h      d.  $10\sqrt{3}$  km/h

2. A ball is dropped from the top of a tower 100m high. Simultaneously another ball is thrown upward with a speed of  $50 \text{ ms}^{-1}$ . After what time do they cross each other?

- a. 1 s      b. 2 s  
c. 3 s      d. 4 s

3. A body is moving with uniform velocity of  $8 \text{ ms}^{-1}$ . When the body just crossed another body, the second one starts and moves with uniform acceleration of  $4 \text{ ms}^{-2}$ . The time after which two bodies meet, will be:

- a. 2 s      b. 4 s  
c. 6 s      d. 8 s

4. A body is projected with a velocity  $u$ . It passes through a certain point above the ground after  $t_1$  sec. The time interval after which the body passes through the same point during the return journey is:

- a.  $\left(\frac{u}{g} - t_1^2\right)$       b.  $2\left(\frac{u}{g} - t_1\right)$   
c.  $3\left(\frac{u^2}{g^2} - t_1\right)$       d.  $3\left(\frac{u^2}{g^2} - t_1\right)$

5. A person standing on an escalator takes time  $t_1$  to reach the top of a tower when the escalator is moving. He takes time  $t_2$  to reach the top of the tower when the escalator is standing. How long will he take if he walks up a moving escalator?

- a.  $t_2 - t_1$       b.  $t_2 + t_1$   
c.  $\frac{t_1 t_2}{(t_1 - t_2)}$       d.  $\frac{t_1 t_2}{(t_1 + t_2)}$

6. A ball is thrown vertically upwards with a speed  $u$ . It reaches a point B at a height  $h$  (lower than the maximum height) after time  $t_1$ . It returns to the ground after time  $t_2$  from the instant it was at B during the upward journey. Then  $t_1 t_2$  is equal to:

- a.  $2h/g$       b.  $h/g$   
c.  $h/2g$       d.  $h/4g$

7. A car starts moving along a line, first with acceleration  $a = 2 \text{ m/s}^2$ , starting from rest then uniformly and finally decelerating at the same rate and comes to rest. The total time of motion 10 s. The average speed during the time is 3.2 m/s. How long does the car moved uniformly?

- a. 4 s      b. 6 s  
c. 5 s      d. 3 s

8. An insect crawls a distance of 4 m along north in 10 seconds and then a distance of 3 m along east in 5 seconds average velocity of the insect is:

- a.  $\frac{7}{15} \text{ m/sec}$       b.  $\frac{1}{5} \text{ m/sec}$   
c.  $\frac{1}{3} \text{ m/sec}$       d.  $\frac{4}{5} \text{ m/sec}$

9. The distance time graph of a particle at time  $t$  makes angle  $45^\circ$  with the time axis. After one second, it makes angle  $60^\circ$  with the time axis. What is the average acceleration of the particle?

- a.  $\sqrt{3} - 1$       b.  $\sqrt{3} + 1$   
c.  $\sqrt{3}$       d. 1

10. An aircraft is flying at a uniform speed  $v \text{ ms}^{-1}$ . If the angle subtended at an observation point on the ground by two positions of the aircraft  $t$  seconds apart is  $\theta$ , the height of the aircraft above the ground is given by (assume that initially aircraft is just above the observer):

- a.  $\frac{vt}{\tan \theta}$       b.  $\frac{2vt}{\tan \theta}$   
c.  $\frac{vt}{\tan\left(\frac{\theta}{2}\right)}$       d.  $\frac{vt}{2 \tan\left(\frac{\theta}{2}\right)}$

11. The velocity of a body moving along a straight line with uniform acceleration  $a$  reduces by  $\frac{3}{4}$  of its initial velocity in time  $t_0$ . The total time of motion of the body till its velocity becomes zero is:

- a.  $\frac{4}{3} t_0$       b.  $\frac{3}{2} t_0$   
c.  $\frac{5}{3} t_0$       d.  $\frac{8}{3} t_0$

12. A train accelerates from rest at a constant rate  $\alpha$  for distance  $x_1$  and time  $t_1$ . After that it retards to rest at constant rate  $\beta$  for distance  $x_2$  and time  $t_2$ . Which of the following relations is correct?

- a.  $\frac{x_1}{x_2} = \frac{\alpha}{\beta} = \frac{t_1}{t_2}$       b.  $\frac{x_1}{x_2} = \frac{\beta}{\alpha} = \frac{t_1}{t_2}$   
c.  $\frac{x_1}{x_2} = \frac{\alpha}{\beta} = \frac{t_2}{t_1}$       d.  $\frac{x_1}{x_2} = \frac{\beta}{\alpha} = \frac{t_2}{t_1}$



13. A train passes an electric post in 8 seconds and a bridge of length 1 km in 108 seconds. The speed of the engine is:

- a.  $10 \text{ ms}^{-1}$       b.  $12 \text{ ms}^{-1}$   
c.  $8 \text{ ms}^{-1}$       d.  $5 \text{ ms}^{-1}$

14. A body starts from the bottom of an inclined plane with angle of inclination  $60^\circ$  with a certain velocity and stops within a distance 10m. If the angle of inclination is reduced to  $30^\circ$  and the body starts with same velocity, it will stop after:

- a. 10 m      b.  $10\sqrt{3}$  m  
c.  $10\sqrt{2}$  m      d. 5 m

15. Rain is falling vertically with a speed of  $20 \text{ ms}^{-1}$ . A person is running in the rain with a velocity of  $5 \text{ ms}^{-1}$  and a wind is also blowing with a speed of  $15 \text{ ms}^{-1}$  (both from the west). The angle with the vertically at which the person should hold his umbrella so that he may not get drenched is:

- a.  $\tan^{-1}\left(\frac{1}{2}\right)$       b.  $\tan^{-1}\left(\frac{1}{3}\right)$   
c.  $\tan^{-1}\left(\frac{4}{3}\right)$       d.  $\tan^{-1}(2)$

16. A boat which has a speed of  $5 \text{ kmh}^{-1}$  in still water crosses a river of width 1 km along the shortest possible path in 15 minute. The velocity of the river water in  $\text{kmh}^{-1}$  is:

- a. 1      b. 3  
c. 4      d.  $\sqrt{41}$

17. A car A is going north east at  $80 \text{ kmh}^{-1}$  and another car B is going south east with a velocity of  $60 \text{ kmh}^{-1}$ . The velocity of A relative of B makes an angle with the north equal to:

- a.  $\tan^{-1}\left(\frac{2}{7}\right)$       b.  $\tan^{-1}\left(\frac{7}{2}\right)$   
c.  $\tan^{-1}(7)$       d.  $\tan^{-1}\left(\frac{1}{7}\right)$

18. A body is thrown vertically upwards from the top A of tower. It reaches the ground in  $t_1$  seconds. If it is thrown vertically downwards from A with the same speed it reaches the ground in  $t_2$  seconds. If it is allowed to fall freely from A, then the time it takes to reach the ground is given by :

- a.  $t = \frac{t_1 + t_2}{2}$       b.  $t = \frac{t_1 - t_2}{2}$   
c.  $t = \sqrt{t_1 t_2}$       d.  $t = \sqrt{\frac{t_1}{t_2}}$

19. The water drops fall at regular intervals from a tap 5 m above the ground. The third drops is leaving the tap at the instant the first drop touches the ground. How far above the ground is the second drop at that instant?

- a. 1.25 m      b. 2.50 m  
c. 3.75 m      d. 4.00 m

20. A man goes at the top of a smooth inclined plane. He releases a bag to fall freely and he himself slides on inclined plane to reach the bottom. If  $v_1$  and  $v_2$  are the velocities of the man and bag respectively, then:

- a.  $v_1 > v_2$   
b.  $v_1 < v_2$   
c.  $v_1 = v_2$   
d.  $v_1$  and  $v_2$  cannot be compared

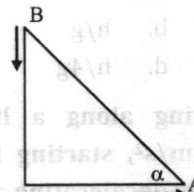
21. If a ball is thrown vertically upwards with speed  $u$ , the distance covered during the last  $t$  seconds of its ascent is:

- a.  $ut - \frac{1}{2}gt^2$       b.  $(u + gt)t$   
c.  $ut$       d.  $\frac{1}{2}gt^2$

22. A boat is sent across a river with a velocity of 8 km/hr. If the resultant velocity of boat is 10 km/hr, then velocity of the river water is:

- a. 12.8 km/hr      b. 6 km/hr  
c. 8 km/hr      d. 10 km/hr

23. Two particles A and B are connected by a rigid rod AB. The rod slides along perpendicular rails as shown here. The velocity of a to the right is 10 m/s. What is the velocity of B when angle  $\alpha = 60^\circ$ ?



- a. 9.8 m/s      b. 10 m/s  
c. 5.8 m/s      d. 17.3 m/s

24. A person swims in a river aiming to reach exactly opposite point on the bank of a river. His speed of swimming is 0.5 m/s at an angle  $120^\circ$  with the direction of flow of water. The speed of water in stream is:

- a. 1.0 m/s                      b. 0.5 m/s  
c. 0.25 m/s                    d. 0.43 m/s

25. A bus starts from rest with an acceleration of 1 m/sec<sup>2</sup>. A man who is 48 meters behind the bus starts with a uniform velocity of 10 m/s. Then the minimum time after which the man will catch the bus:

- a. 4 sec                      b. 10 sec  
c. 12 sec                    d. 8 sec

26. A body starts from rest with an acceleration  $a_1$ . After two seconds another body B starts from rest with an acceleration  $a_2$ . If they travel equal distances in fifth second after the start of A, the ratio  $a_1 : a_2$  will be equal to:

- a. 9 : 5                      b. 5 : 7  
c. 5 : 9                      d. 7 : 9

27. A point traversed half of the distance with a velocity  $v_0$ . The remaining part  $v_1$  for half of the time and with the velocity  $v_2$  for the other half of the time. The mean velocity of the point over the whole time of motion will be:

- a.  $\frac{2v_0(v_1 + v_2)}{(2v_0 + v_1 + v_2)}$       b.  $\frac{v_0 + 2v_1 + 2v_2}{3}$   
c.  $\frac{2v_0 + v_1 + v_2}{3}$                       d.  $\frac{v_0 + v_1 + v_2}{3}$

28. A car moving with a speed of 40 km/hr can be stopped by applying brakes after at least 2 metres. If the same car is moving with a speed of 80 km/hr. Then the minimum stopping distance will be:

- a. 2m                      b. 4m  
c. 8m                      d. 6m

29. A body covers 200 cm in the first two seconds and 220 cm in the next 4 seconds. The velocity of the body at the end of the 7th second will be (assume acceleration to be constant):

- a. 30 cm/sec                      b. 25 cm/sec  
c. 10 cm/sec                      d. 15 cm/sec

30. A elevator car whose floor to ceiling distance is 2.7 m starts ascending with a constant acceleration of 1.2 m/s<sup>2</sup>, 2 second after the start a bolt falls from the ceiling of the car. The free fall time of the bolt is: ( $g = 9.8 \text{ m/s}^2$ )

- a.  $\sqrt{\frac{2.7}{9.8}} \text{ s}$                       b.  $\sqrt{\frac{5.4}{9.8}} \text{ s}$   
c.  $\sqrt{\frac{5.4}{8.6}} \text{ s}$                       d.  $\sqrt{\frac{5.4}{11}} \text{ s}$

31. Two trains, each of 50 m length are moving parallel on parallel lines towards each other with speeds of 10 m/s and 15 m/s. They pass each other in:

- a. 6 sec                      b. 2 sec  
c. 4 sec                      d.  $5\sqrt{\frac{2}{3}} \text{ sec}$

32. The engineer of a train moving at a speed  $v_1$  sights a freight train a distance  $d$  ahead of him on the same track moving in the same direction with a slow speed  $v_2$ . He puts on the brakes and gives his train a constant deceleration  $\alpha$ . Then there will be no collision if:

- a.  $d < \left(\frac{v_1 - v_2}{2\alpha}\right)$                       b.  $d < \left(\frac{v_1 - v_2}{2\alpha}\right)^2$   
c.  $d > \frac{(v_1 - v_2)^2}{2\alpha}$                       d.  $d = \frac{v_1 - v_2}{\alpha}$

33. A boat is moving with a velocity of  $3\hat{i} + 4\hat{j}$  and water in river is moving with a velocity  $-3\hat{i} - 4\hat{j}$  with respect to ground. The relative velocity of boat with respect to water is:

- a.  $8\hat{j}$                       b.  $-6\hat{i} - 8\hat{j}$   
c.  $6\hat{i} + 8\hat{j}$                       d.  $6\hat{i}$

34. A car accelerates from rest at a constant rate  $\alpha$  for some time, after which it decelerates at a constant rate  $\beta$  and comes to rest. When total elapsed time is  $t$ , then the maximum velocity acquired by the car is:

- a.  $\frac{(\alpha^2 + \beta^2)t}{\alpha\beta}$                       b.  $\frac{(\alpha^2 - \beta^2)t}{\alpha\beta}$   
c.  $\frac{(\alpha + \beta)t}{\alpha\beta}$                       d.  $\frac{\alpha\beta t}{\alpha + \beta}$

### Answers:

|       |       |       |       |       |
|-------|-------|-------|-------|-------|
| 1. c  | 2. b  | 3. b  | 4. b  | 5. d  |
| 6. a  | 7. b  | 8. c  | 9. a  | 10. a |
| 11. a | 12. b | 13. a | 14. b | 15. a |
| 16. b | 17. d | 18. c | 19. c | 20. c |
| 21. c | 22. b | 23. c | 24. c | 25. d |
| 26. c | 27. a | 28. c | 29. c | 30. d |
| 31. c | 32. c | 33. c | 34. d |       |

### Solution for Selected Problems

1.  $v_m = 12 \text{ km/hr}$   $v_r = ?$

Angle with vertical i.e., angle with  $\vec{v_r}$  is  $= 60^\circ$

$$\tan 60^\circ = \frac{v_m}{v_r}$$

$$v_r = \frac{12}{\sqrt{3}} = 4\sqrt{3} \text{ km/hr}$$

3. Let two bodies meet after time 't' then

$$8t = \frac{1}{2} \times 4 \times t^2$$

$$\therefore t = 4 \text{ sec}$$

4. At time ' $t_1$ ' height is given by

$$h = ut_1 - \frac{1}{2}gt_1^2 \dots\dots\dots(i)$$

$$\text{at } t = t_2, h = ut_2 - \frac{1}{2}gt_2^2 \dots\dots\dots(ii)$$

Solving (i) & (ii) we get,

$$h = \frac{1}{2}g(t_1 + t_2)$$

$$\therefore t_2 = \frac{2u}{g} - t_1$$

$$\text{time difference, } \Delta t = t_2 - t_1 = 2\left(\frac{u}{g} - t_1\right)$$

5. Speed of escalator ( $v_e$ ) =  $\frac{h}{t_1}$ ; where h = height of tower

$$\text{speed of man } (v_m) = \frac{h}{t_2}$$

$$\text{time taken when both moves } t = \frac{h}{v_e + v_m} = \frac{h}{\left(\frac{h}{t_1} + \frac{h}{t_2}\right)}$$

$$\therefore t = \frac{t_1 t_2}{t_1 + t_2}$$

7.  $\therefore$  time for accelerate from rest and retards wifer same route and comes to rest are sane i.e., t (say)

Net time for uniformly moves =  $t_1$

$\therefore$  total time of motion

$$2t + t_1 = 10 \dots\dots\dots(i)$$

$$\text{Average speed} = \frac{\text{total distance}}{\text{total time}}$$

$$\Rightarrow 3.2 = \frac{\frac{1}{2}at^2 + at \times t_1 + \frac{1}{2}at^2}{10}$$

$$\therefore t^2 + t t_1 = 16 \dots\dots\dots(ii)$$

Solving equation (i) and (ii) we get

$$t = 8 \text{ sec} \& 2 \text{ sec}$$

But  $t = 8 \text{ sec}$  is not possible

$$\therefore t = 2$$

$$\text{from (1) } t_1 = 6 \text{ sec}$$

$$8. \text{ Average velocity} = \frac{\text{total displacement}}{\text{total time}} = \frac{\sqrt{4^2 + 3^2}}{(10 + 5)}$$

$$= \frac{5}{15} = \frac{1}{3} \text{ m/sec}$$

$$9. v_1 = \tan 45^\circ = 1$$

$$v_2 = \tan 60^\circ = \sqrt{3}$$

$$\text{Average acceleration} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\sqrt{3} - 1}{1} = \sqrt{3} - 1$$

$$10. \tan \theta = \frac{vt}{h} \Rightarrow h = \frac{vt}{\tan \theta}$$

11. Let  $V_0$  be the initial velocity,

$$v = u + at$$

$$\Rightarrow \frac{1}{4}V_0 = V_0 + at_0$$

$$a = -\frac{3V_0}{4t_0}$$

$$\text{Again, } v = u + at$$

$$0 = V_0 + \left(-\frac{3V_0}{4t_0}\right) \times t$$

$$\therefore t = \frac{4}{3}t_0$$

$$12. V = \alpha t_1 = \beta t_2 \Rightarrow \frac{\alpha}{\beta} = \frac{t_2}{t_1}$$

$$\text{again, } V^2 = 2\alpha x_1 = 2\beta x_2 \Rightarrow \frac{\alpha}{\beta} = \frac{x_2}{x_1}$$

$$\therefore \frac{\alpha}{\beta} = \frac{x_2}{x_1} = \frac{t_2}{t_1}$$

$$\therefore \frac{x_1}{x_2} = \frac{\beta}{\alpha} = \frac{t_1}{t_2}$$

$$14. \therefore 0^2 = u^2 - 2 \times g \sin 60^\circ \times 10$$

$$\Rightarrow u^2 = 2 \times 10 \times \frac{\sqrt{3}}{2} \times 10 = 100\sqrt{3}$$

Again,